

4	(a)	(i)	$n = \frac{N}{\text{Volume}} = \frac{3.2 \times 10^{22}}{3.0(1.3 \times 10^{-7})} = 8.2 \times 10^{28}$	B1
		(ii)	$I = nAve$ $v = \frac{I}{nAe} = \frac{0.80}{8.2 \times 10^{28}(1.3 \times 10^{-7})(1.6 \times 10^{-19})}$ $= 4.7 \times 10^{-4} \text{ m s}^{-1}$	M1 A1
	(b)	(i)	$V_w = IR = (0.80)(0.40) = 0.32 \text{ V}$ C1 Potential divider $V_w = \frac{R}{R+r} E$ $0.32 = \frac{0.40}{0.40+r} (0.48)$ $r = 0.20 \Omega$ A1	
		(ii)	$\% \text{ loss} = \frac{I^2 r}{I^2 (r + R)} = \frac{0.20}{0.60} = 0.33 = 33\%$	B1
	(c)	(i)	Wire Y has <u>smaller resistance</u> Overall <u>current</u> in circuit <u>increases</u> Drift velocity is greater.	M1 A1
		(ii)	$\% \text{ loss} = \frac{I^2 r}{I^2 (r + R)} = \frac{r}{(r + R)}$ which is the ratio of internal resistance to total resistance. % losses increases when R decreases, efficiency decreases when wire Y is used *ecf if (c)(i) R increases	M1 A1

5	(a)	(i)	$P = \mu nI$	
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